Freight Audit and Rate Intelligence Engine

Build roadmap, completed functions and release gates

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Current position: R1 and R2 completed, tested locally, committed and pushed.
Next milestone: Weight validation, followed by contract pricing.

Release objective

Build a carrier configurable parcel audit engine that can ingest supported invoice layouts, validate supported charges, explain findings and produce traceable reports. Carrier independence is the design direction, not a claim that every carrier or freight mode is already supported.

Release track	Purpose	Position
Core engine V1	Reliable invoice audit, evidence review and report export for explicitly supported carrier configurations.	R1 and R2 complete; R3 to R7 planned.
Website launch	A controlled user interface and hosted service around the validated engine.	W1 and W2 planned after the engine pilot.
Later extensions	Rate comparison, historical intelligence, integrations and additional freight modes.	Backlog, outside core V1.

How to read this roadmap

Completed items reflect the project files and confirmations in this conversation. The R3 to R7 breakdown below is a consolidated forward plan, not a claim that the exact numbering has been recovered from the earlier design PDF. Website and extension phases are shown separately to keep the V1 scope controlled.

No completion percentage or launch date is assigned. Remaining calculation, evidence and release work must pass its own acceptance tests before a launch commitment is credible.

What has been achieved

The current build is a working audit foundation, not yet the complete parcel audit product.

Phase	Functions delivered	Result
Initial engine	Invoice loading, repeated AWB diagnostics, basic carrier and service rate checks, exposure reconciliation and Excel reporting.	Working baseline retained.
R1: Standard data and carrier configuration	Standard invoice fields; carrier mapping files; carrier profiles; configuration validation; command line carrier selection.	Carrier layouts and settings can be configured without embedding each carrier in the core logic.
R1: Traceability	Rule catalogue, rule IDs and versions; unique audit run IDs; timestamps; input hashes; processing status; failure records; report linkage.	Findings can be traced to the rules and run that produced them.
R2: Charge matching	Separate charge types; exact positive charge duplicate matching; labelled legacy matching when charge type is absent.	Freight and fuel sharing an AWB are not automatically treated as duplicate charges.
R2: Credits and reporting	Conservative full reversal matching; linked source lines; review treatment for unclear credits; rate scope safeguards; expanded reports and run summaries.	Matched reversal pairs are excluded from exposure; ambiguous credits remain for review.

Validation and repository checkpoint

- The final corrected R2 patch passed the check against the uploaded current files.
- The reconstructed test environment reported 268 passed and 2 skipped. The skipped tests depended on the unavailable workbook.
- You confirmed that the local installation and all local tests passed. The exact local test count was not restated.
- You confirmed the commit and push commands worked. Patch and temporary transfer files were excluded from the intended commit.

What the engine does today

These boundaries must remain visible in demonstrations, reports and future product claims.

Current end to end workflow

Load invoice and optional rate data → apply carrier mapping and configuration → identify repeated AWBs and charge duplicates → classify credits and reversals → perform current freight rate checks → reconcile quantified exposure → export the report and run log.

R2 matching rules

- A typed duplicate uses carrier, AWB, currency, charge type and exact positive amount. Subsequent matching active charges are flagged.
- A full reversal requires a unique positive and negative pair with complete matching identity and equal absolute amount. Invoice reference, service and charge components must agree when present.
- Partial, excess and ambiguous credits are not allocated automatically. Matching is within the input file, not across invoice history.
- Explicit nonfreight lines are not compared with freight rates. Missing or unsupported charge information is not treated as proof of a valid charge.

Important limitations still open

- Current rate lookup is based on carrier and service, not complete zone, weight, currency and effective date pricing.
- Carrier profile fields for dimensional weight and rounding do not mean weight validation has been implemented.
- Delivery evidence, accessorial rates and tax calculations are not yet validated.
- Use one verified currency per run. Mixed currency aggregation and currency conversion are not supported.
- Unknown exposure is not zero liability. Quantified totals are not automatically approved savings or payment instructions.

Control principle

Keep the original invoice data unchanged. Record findings separately, retain the reason for each review outcome and require human approval for payment, recovery or dispute action.

R3 | PLANNED

Weight and measurement validation

Calculate the correct chargeable weight before selecting the freight rate.

Functions to build

- Validate actual weight, package dimensions, units and package count. Distinguish missing values from zero values.
- Convert supported units consistently while preserving the original invoice values and unit labels.
- Calculate dimensional weight using the configured divisor and measurement basis. Apply the supported carrier rounding rules.
- Compare expected chargeable weight with billed weight and classify differences using configured tolerances.
- Define package level and shipment level aggregation explicitly. Unsupported multi package structures remain for review.

Deliverables

- A weight calculation module with calculation evidence, configuration validation and rule tagged findings.
- Tests for actual versus dimensional weight, rounding boundaries, unit conversion, missing dimensions and multi package limits.

Completion gate

Known examples reproduce independently checked weights. Invalid or incomplete measurements cannot silently produce a passing result. Existing tests remain passing.

Dependency and boundary

Build on R1 carrier profiles and standard fields. This phase validates weight; it does not yet provide the complete zone and weight rate calculation.

R4 | PLANNED

Contract pricing and rate validation

Replace the basic lookup with supported, date aware parcel pricing.

Functions to build

- Validate rate cards and reject overlapping or ambiguous rate entries rather than selecting an arbitrary match.
- Resolve supported origin and destination zones using carrier specific mapping data. Missing zones remain explicit review items.
- Select the rate by carrier, service, currency, zone, expected chargeable weight band and shipment date.
- Support clearly declared pricing methods, starting with the parcel structures required by the pilot carriers. Keep other freight modes outside this release.
- Compare billed freight with expected freight and record the selected rate, contract version, date basis and calculation steps.

Deliverables

- Rate card schema, zone resolver, supported pricing methods and a traceable freight calculation result.
- Tests for weight band edges, rate validity dates, zone gaps, duplicate rate entries, currency mismatch and missing rates.

Completion gate

Supported pilot shipments reproduce agreed contract examples. Expired, ambiguous or incompatible rates never produce a false pass. Unknown amounts remain distinct from quantified overcharges.

Dependency and boundary

Requires R3 expected chargeable weight. Retain the old lookup only as an explicitly labelled compatibility mode, not as the full V1 pricing claim.

R5 | PLANNED

Delivery evidence and shipment status

Connect invoice charges to delivery evidence without confusing missing evidence with confirmed nondelivery.

Functions to build

- Accept supported delivery evidence files and map them to a standard record containing carrier, AWB, event date, status and source reference.
- Normalize delivery statuses and match evidence conservatively to the invoiced shipment. Detect missing, conflicting or ambiguous evidence.
- Distinguish delivered, confirmed exception, returned, cancelled and evidence unavailable outcomes where supported by the input.
- Apply configurable review policy to unresolved delivery status. Missing evidence must not be described as proof that delivery failed.
- Carry evidence references and match reasons into the report and run log without exposing unnecessary personal data.

Deliverables

- Delivery evidence loader, matching rules, status classification and report fields.
- Tests for valid delivery, missing evidence, conflicting events, duplicate AWBs and unsupported statuses.

Completion gate

Each delivery outcome is traceable to its source. Unmatched or conflicting evidence is visible for review, and no shipment is cleared merely because a record exists.

Dependency and boundary

Uses the standard invoice identity and R1 traceability. Start with file based evidence; live carrier tracking integrations are later extensions.

R6 | PLANNED

Surcharges, tax and final audit decisions

Complete the supported charge calculations and combine findings into explainable review outcomes.

Functions to build

- Validate supported fuel and accessorial charges using configured charge codes, effective dates, calculation bases and contract rules.
- Recalculate supported tax amounts from an explicitly configured taxable base and rounding method. Tax applicability requires verified configuration, not inferred legal advice.
- Check charge components against invoice totals and expose inconsistent totals, missing bases and unsupported charge types.
- Combine duplicate, credit, weight, freight, delivery, surcharge and tax findings using a documented decision precedence.
- Prevent overlapping findings from double counting the same recoverable amount. Preserve unknown amounts and separate review outcomes from payment approval.

Deliverables

- Surcharge and tax modules, total reconciliation, decision policy and a documented exposure allocation method.
- Integrated tests for multiple findings on one line, reversal pairs, component discrepancies and unknown exposure.

Completion gate

Every final outcome has rule IDs and supporting calculations. Exposure reconciles to its components without double counting. Unsupported tax or surcharge rules remain unassessed, not valid.

Dependency and boundary

Requires R2 to R5. Freeze the final decision labels and precedence here. The existing report labels are not yet a complete V1 decision framework.

R7 | PLANNED

Engine V1 pilot and release

Turn the completed calculation engine into a reproducible, documented release.

Functions to build

- Freeze the supported carrier configurations, invoice layouts, currencies, pricing methods and evidence formats for V1.
- Run end to end acceptance tests using independently checked examples and representative permitted data. Compare the private workbook baseline where available.
- Exercise at least three distinct carrier configurations through mappings and profiles, without carrier specific core code changes.
- Check malformed files, failure recovery, deterministic calculations, input immutability, report reconciliation and performance on declared file sizes.
- Complete setup instructions, sample data, configuration guidance, rule documentation, limitations, troubleshooting and release notes.
- Run a controlled pilot with manual review, record defects, resolve release blockers and tag the approved repository release.

Deliverables

- Versioned engine release, acceptance evidence, pilot defect log, supported configuration list and operating guide.
- Repeatable installation and test process with private data, credentials and generated outputs excluded from the public release.

Completion gate

All required regression and acceptance tests pass; workbook tests are run where required rather than silently skipped; no unresolved critical defects remain. Pilot results are approved against the declared scope.

Dependency and boundary

This gate releases the engine, not a public website. Reconfirm the earlier 1,240 line workbook target against the actual available workbook and approved expected results before using it as a release criterion.

From engine to website launch

The website is a separate delivery track. It should call the same tested engine rather than recreate the audit rules.

W1: Controlled web interface

- Build the invoice upload, carrier selection, rate card selection and optional delivery evidence upload screens. Validate required inputs before starting a run.
- Show processing state, run history, summary results, review filters, line level evidence and downloadable reports.
- Display unknown exposure, unsupported rules and configuration limitations prominently. Do not present potential exposure as confirmed savings.
- Add authenticated access, separation between users or organisations, file retention controls and an audit trail appropriate to the pilot.

Completion gate: A permitted user can complete a full audit through the interface, retrieve the correct report and access only authorised runs. Interface totals agree with the engine output.

W2: Hosted pilot and public launch readiness

- Set up the hosting environment, secure configuration, file storage, background processing, monitoring and backup and restore procedures.
- Define file size limits, timeouts, retention and deletion behaviour. Test access controls, upload validation and failure handling.
- Complete user guidance, privacy and service documentation, support process and incident response ownership. Obtain appropriate review before accepting customer data.
- Run a restricted hosted pilot, resolve launch blockers and confirm the release scope before opening access more widely.

Completion gate: Hosting, security, recovery, privacy and user acceptance checks are signed off for the intended audience. Launch only the carrier configurations and checks that have passed acceptance.

Boundary: Subscription billing, enterprise identity integrations and commercial service commitments are separate decisions. They are not assumed complete or mandatory for a private portfolio demonstration.

Beyond Version 1

These phases preserve the wider vision without delaying the core parcel audit release.

Phase	Coverage	Dependency
B1: Audit history	Persistent invoice history, cross file duplicate detection, cross invoice credit matching and controlled partial credit allocation.	Stable identifiers, storage design and approved matching policy.
B2: Rate intelligence	Price the same shipment across supported carriers; compare route, zone, service and contract date; explain price and service tradeoffs.	Validated R3 and R4 pricing plus comparable, authorised carrier rate data.
B3: Operational intelligence	Trends, recurring billing issues, carrier scorecards, recovery case tracking and evidence packs.	Reliable historical data and human approved recovery outcomes.
B4: Integrations	Carrier tracking feeds, ERP and transport systems, scheduled ingestion and approved export workflows.	Stable interfaces, access controls and operational ownership.
B5: Broader transport coverage	Additional pricing methods for air, ocean, road freight and other supported modes; broader currency reporting where required.	Mode specific schemas, contracts, calculation modules and independent acceptance tests.

Rule for admitting a new feature

Build it in V1 only if it is required to audit the declared parcel scope correctly. Capture fields now if adding them later would force a data model rewrite. Otherwise keep the feature in the extension backlog.

End of the current project scope

R7 closes the core engine build. W2 closes the website launch track if that track is pursued. B1 to B5 are subsequent product releases, not unfinished obligations that must all be completed before V1 can launch.

How we finish without losing progress

Keep each phase small enough to verify, transfer and commit safely.

Immediate next steps

- Preserve the committed R2 checkpoint and confirm the repository working state before beginning new work.
- Refresh the project bundle from the current local code. Use exact source files when preparing patches, preserving file endings.
- Agree the first pilot carrier configurations and obtain verified dimensional rules, rounding rules, rate cards and worked examples.
- Start R3 with tests for expected chargeable weight. Complete and commit it before moving into R4 pricing.

Repeat for every phase

Confirm baseline → define acceptance examples → write tests → implement the smallest complete change → run regression tests → verify the patch on exact source files → install locally → run local tests → review reports → commit and push → update this roadmap.

V1 release checklist

- Supported invoice inputs and carrier configurations are declared and validated.
- Weight, freight, delivery evidence, supported surcharges and supported tax checks have passed acceptance.
- Duplicate and credit handling remain conservative and traceable.
- Reports reconcile, unknowns remain visible and overlapping findings do not inflate exposure.
- The full required test suite and controlled pilot pass with documented limitations.
- Installation, configuration, data handling and support instructions are ready.
- If launching online, W1 and W2 access, security and recovery gates also pass.

Basis and status notes

Source basis: the project bundle and R2 patch supplied in this conversation; the exact audit and exposure source files; the recorded test outputs; and your confirmations that local tests, commit and push succeeded. Future phases are planned scope, not implemented functions. No external market, legal or pricing assumptions are used.

This roadmap deliberately avoids a speculative completion percentage or calendar estimate. Reassess delivery effort after the R3 and R4 pilot examples establish the actual carrier and pricing complexity.